LLMs as boundary phenomena: Bearers and cognitive attribution

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Abstract

Attributions of cognition to large language models often shift among four bearers: a base checkpoint, a prompt-bounded session, a deployed system, and an agentic scaffold. That shift matters for Nefdt (2026)'s claim that LLMs occupy the language-without-cognition quadrant because they're unplugged from wider cognitive structure and the world. The argument fits text-only checkpoints and sessions, but it doesn't automatically transfer to scaffolds with tools, persistent memory, sensors, and maintained goals. Here, PROJECTIBILITY means licensed reach beyond the cases in hand. A PROJECTION is an inference from an observed source to an unobserved target; its declaration specifies the bearer, population, conditions, transformations, timescale, warrant, tolerance, and failure condition. The schema distinguishes a failed theory-of-mind transfer, a bounded intervention-backed organizational projection, and a failed cross-construction processing projection. It also clarifies how broader evidence for cognitive convergence should be tested. A property-cluster account supplies a prospective hypothesis: properties that co-cluster in humans may dissociate in systems produced and integrated differently. Any defensible answer about LLM cognition has to be indexed to a bearer and to the projection at issue.

Keywords: large language models; cognition; projectibility; model evaluation; natural kinds; agency

1 Introduction

A claim about what an LLM can do is incomplete until we know which LLM-like bearer it concerns. A fixed base checkpoint, one prompt-bounded session, a deployed chat system, and an agentic scaffold can differ in memory, tools, sensors, control loops, and goal persistence. Evidence about one needn't license an expectation about the others.

That distinction creates a first-order problem for Nefdt (2026). Nefdt argues that current LLMs occupy a previously missing quadrant: language without cognition. They're "purely linguistic agents unplugged from integration with both larger cognitive structure and the world in which it evolved"

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(sec. 4.3). The description is plausible for some base checkpoints and text-only sessions. It doesn't follow for every deployed system or scaffold assembled around them.

The choice is substantive. If the missing-quadrant claim concerns only checkpoints or chat-only sessions, its bearer is narrower than the systems now at issue in many debates about AI agency. If it extends to scaffolds with tools, persistent memory, sensors, and maintained goals, it becomes an empirical claim about integration. Such a claim can fail. Adding these resources doesn't establish cognition, but it blocks an inference from the limitations of one bearer to the absence of integration in another.

This paper uses PROJECTIBILITY (what observing one feature licenses us to expect about another; Goodman, 1955/1983) to expose and repair that inference. A projective claim treats an observed source as a reason to expect an unobserved target for a specified bearer and population under declared conditions. Category membership is a downstream verdict about a family of such claims. It can't substitute for them.

The bearer-relative objection shows where Nefdt's unrestricted placement outruns its evidence. A declaration-and-adjudication schema then distinguishes failed, bounded, and currently untestable projections. Cases from theory-of-mind evaluation, mechanistic intervention, and psycholinguistic transfer show the schema doing more than recommending caution. Finally, a property-cluster account links reliable induction to causal structures that sustain imperfect clusters and supplies a prospective hypothesis about where cognitive properties should dissociate.

2 Projectibility and its neighbours

The term PROJECTIBILITY has a narrower philosophical lineage than its use here. Goodman's problem concerns which predicates support induction rather than merely fit the observations (Goodman, 1955/1983). Boyd connects that epistemic role to classificatory practices accommodated to causal structure (Boyd, 1991, 1999). Here, projectibility means *licensed ampliative reach*: what an observation warrants us in expecting beyond the cases used to establish it, under which scope conditions, and with what failure rule. The usage inherits the inductive question without importing Goodman's full account of entrenchment.

The question also resembles Dennett's intentional stance. On a stance-based approach, intentional interpretation is justified when it supports successful prediction of a system's behaviour (Dennett, 1987, 1991). Nefdt himself invokes real patterns and intentional-stance instrumentalism when locating his view between realism and eliminativism (sec. 4.3). Projectibility adds a different grain of analysis. It decomposes a whole-system stance into source–target claims that can succeed or fail separately, and it requires the bearer and stopping point to be fixed before the inference is assessed.

Shanahan (2024) similarly warns that predicates such as *knows*, *believes*, and *thinks* import expectations from human cases. The projective schema turns that warning into a test. It identifies which expectations exceed the evidence while allowing other uses of the same predicate to remain licensed.

The proposal likewise neighbours measurement validity. Raji et al. (2021) argue that benchmarks are finite and contextual. Performance doesn't automatically measure the general capabilities for which a benchmark is used as a stand-in. Harding and Sharadin (2024) ask what it's for a model to possess a capability and analyse ability in terms of reliable success under appropriate background conditions. The schema extends this discipline from instruments and ability claims to cognitive predicates. It asks which further expectations an attribution licenses across tasks, levels, times, and bearers.

Projectibility is a diagnostic for ampliative inference, not a complete theory of cognition. The position is pluralist because cognitive capacities, organization, memory, agency, embodiment, and phenomenology may dissociate. It's contextualist only in the limited sense that an inquiry selects a target. Purpose determines which projection is at issue; evidence determines whether it succeeds. A useful attribution isn't thereby metaphysically fundamental, naturally kinded, or phenomenally real.

3 The bearer-relative objection

Nefdt's table is best read charitably as a heuristic, not as a commitment to binary essences. His argument supplies a route to the language-without-cognition cell. The NO BRAINER principle says that LLMs model one aspect of cognition, statistical linguistic processing. The COGNITION UNPLUGGED argument adds that their intentional states are proxies disconnected from wider cognition and the world (sec. 3.2). The source evidence concerns architecture, training, and text-mediated performance; the target is the absence of broader integration.

The route crosses bearers. Table 1 distinguishes four objects routinely called *an LLM*. They needn't support the same projections.

Bearer	Included resources	What remains open
Base checkpoint	Architecture and fixed learned parameters	Behaviour under prompting, tools, or persistent state
Prompt-bounded session	Checkpoint plus the current context and in-context state	Transfer across sessions and persistence beyond the context
Deployed system	Session plus interface, policies, retrieval, and selected tools	Which resources are integrated into control and action
Agentic scaffold	Deployed system plus multi-step control; potentially memory, sensors, actuators, and maintained goal state	Robustness, autonomy, and the extent of cross-resource integration

Table 1: Four bearers commonly conflated in cognitive attributions to LLMs.

The strongest charitable declaration of Nefdt's inference is correspondingly narrow. Its bearers are text-only base checkpoints and prompt-bounded sessions. Its source is their training and architecture together with successful linguistic proxies. Its target is the absence of integration with non-linguistic information, persistent memory, and world-involving action. Its conditions exclude tools, cross-session state, sensors, and independent goal maintenance. Direct architectural inspection and the comparison with a selectively dissociable human language system provide the warrant. Repeated, intervention-sensitive use of non-linguistic input and persistent goals across tasks by a richer bearer would defeat the broader projection.

That claim may be defensible. It doesn't project unchanged to the other rows of Table 1. Nefdt acknowledges that retrieval-augmented generation connects a model to online sources, but treats that interaction as still linguistic and pattern-based (sec. 3.2, n. 13). Retrieval alone may indeed leave the deeper issue untouched. A scaffold that combines retrieval with persistent state, multimodal sensors, action, feedback, and maintained goals presents a different bearer. Whether those resources form wider cognitive integration is an empirical question, not something settled by the checkpoint's training objective.

Nefdt's options now diverge. The missing-quadrant verdict can be restricted to a bearer for which the unplugging argument is supported. Alternatively, it can range over deployed and scaffolded systems, provided the absence of integration is stated as a defeasible projection and tested there. The unrestricted claim that LLMs as such occupy the quadrant isn't licensed. Indexing the bearer changes the verdict.

Other targets require separate treatment. Embodiment, persistent memory, and goal maintenance are inspectable features whose presence depends partly on deployment. Phenomenology isn't another resource on that list. Without a bridge principle and an accessible test, architectural description underdetermines what, if anything, the system experiences. The projection from architecture to experience is unlicensed rather than empirically defeated; the same evidence supports neither presence nor absence.

4 Declaration before adjudication

A projective claim should separate what's fixed before the test from what's decided afterwards. Table 2 preserves that temporal order. It isn't a theory of cognition or a demand that every article print a form. It states the information needed to tell whether a result has travelled farther than its evidence.

Stage	Field	Question fixed
Declaration	Bearer and population	Which checkpoint, session, deployment, or scaffold; which sampled and target systems?
Declaration	Source and target	Which observation starts the inference, and which independent outcome is expected?
Declaration	Conditions, transformations, and timescale	Under which task and context, across which task-preserving changes, and for how long?
Declaration	Warrant	Which evidence and bridge principle license the expectation?
Declaration	Tolerance and failure condition	How much variation counts as success, and what prospective result would defeat the claim?
Adjudication	Result	Did the declared target meet the tolerance?
Adjudication	Transfer limit	Where does the inference stop across tasks, levels, times, or bearers?
Adjudication	Revision	Should the claim be broadened, restricted, split, suspended, or retired?

Table 2: Prospective declaration and post-result adjudication of a projective claim.

The distinction prevents two common repairs from becoming invisible. First, a failed projection can't be saved by redefining the population after the misses are known. Secondly, a successful result can't move from a task to a capacity, or from a session to a scaffold, without an additional bridge principle. The same observation can support a local attribution while failing to support a broader one.

5 What the empirical cases establish

5.1 A failed theory-of-mind transfer

Kosinski (2024) tested eleven LLMs on 640 prompts distributed across forty false-belief tasks and matched controls. GPT-4 solved 75% of the tasks, and the article raises the possibility that theory of mind emerged as a by-product of improving language skills. That interpretation is cautious, but it still identifies a candidate projection: from success on the battery to a capacity for attributing false beliefs under relevantly varied conditions.

The skipped field is the transformation regime. The bearer is a set of specific historical model endpoints, and the source is success on the constructed prompts. A broader theory-of-mind target requires stability under changes that preserve the false-belief problem while altering incidental wording, objects, or expectations. Ullman (2023) reports that small task-preserving alterations reverse the apparent success. That result doesn't erase performance on Kosinski's battery. It defeats the broader transfer until a prospective tolerance is met across such variations.

The schema changes what a reader is entitled to expect. The supported claim is that specified model versions solved specified false-belief tasks under specified prompting. The failed projection is that this success licenses robust theory-of-mind performance across nearby realizations. Neither verdict settles phenomenal perspective, human-like social cognition, or the performance of a scaffold whose memory and interaction history differ from those model endpoints.

5.2 A bounded organizational success

Hu et al. (2026) provide a positive case. Seven tested checkpoints show a default same-colour tendency in a verbal crayon task, and six show a congruency effect. For Gemma-2-2B, attribution and attention evidence identify competition between an in-weight default and an in-context rule. Attention ablation, fine-tuning, and rule-set manipulations then differentially affect congruent and incongruent trials in the predicted direction. Related manipulations across six Pythia checkpoints produce broadly similar patterns, although Pythia-2.8B lacks the behavioural congruency effect.

Here the source isn't behavioural resemblance alone. Convergent interventions warrant a narrow target: a causally implicated processing route for conflict resolution in the tested task and models. The bearer and method matter because causal attribution graphs were available only for Gemma, whereas the broader Pythia comparison used the other analyses. The claim shouldn't silently become a shared human mechanism or a property of all LLMs.

A broader projection would require the directional intervention pattern to survive prespecified paraphrases, new conflict tasks, and new model families. Loss or reversal under those transformations, or failure of a pathway-specific intervention, would defeat it. This prospective rule makes the organizational attribution informative without converting it into general cognition.

5.3 A failed processing projection

von der Malsburg and Padó (2026) show why success at one construction doesn't establish a human-like processing organization. Eleven autoregressive transformers produce generally human-aligned surprisal patterns for agreement attraction in prepositional-phrase modifiers. The target is recurrence of the human singular–plural and grammaticality asymmetries in object-extracted relative clauses under the same processing account.

The target fails: no model reproduces the human pattern in the relative-clause condition, and predictions diverge markedly across models. The proper revision is construction-bounded. The result doesn't show that transformers lack grammatical sensitivity, nor that no model can approximate human processing. It splits sensitivity to agreement conditions from projection of a stable human processing signature. Broadening would require prospectively declared success across adjacent constructions, languages, and model families.

The same discipline clarifies adjacent learnability evidence. Kallini et al. (2024) train GPT-2-small models on English-derived counterfactual corpora and find greater difficulty for typologically impossible than possible transformations. The result bears on controlled learnability and inductive bias. It doesn't project directly to nearby-task performance by a deployed conversational system, much less to its memory, agency, or phenomenology.

6 The convergence challenge

Sripada and Lewis (2026) present the strongest challenge to decomposition. They synthesize evidence for correspondences in inferential organization, computational architecture, representational structure, prediction-driven learning, and reinforcement-learning-like mechanisms for goal-directed action. If those properties co-vary tightly across model families, tasks, and bearer changes, then separating them would become idle bookkeeping and a broader cognitive attribution would gain support.

The current evidence doesn't yet establish that joint pattern. Its components come from studies with different models, tasks, interventions, and levels of description. A representational correspondence needn't travel with persistent memory; a conflict-processing route needn't travel with autonomous goal maintenance; a learning analogy needn't establish a shared biological implementation. Convergence at several targets hasn't established convergence of the targets in the same bearers.

The synthesis also proposes a research programme for broader projection. The appropriate test would sample the same declared bearers across capacities, organization, memory, and goal-directed control, then examine whether the pattern survives task-preserving variation and changes of model family and deployment. Consistent co-variation would weaken the boundary argument. Stable dissociations would strengthen it. The schema adjudicates the disagreement by specifying what evidence could move either side.

7 Property clusters as a prospective hypothesis

Homeostatic property cluster (HPC) theory explains some reliable projections through imperfect property clusters sustained by causal structures (Boyd, 1991, 1999). That explanatory role follows the projective declaration rather than replacing it. A plausible mechanism doesn't by itself fix the bearer, target, conditions, or tolerance, and a warranted projection needn't yet have a known causal ground.

HPC theory can guide inquiry prospectively. Capacities, computational organization, embodiment, memory, agency, and phenomenology often travel together in humans. If common developmental, embodied, and evolutionary structures sustain that pattern, systems produced and assembled through different pathways should split some of its components. Powell (2020)'s distinction between convergent and contingent properties sharpens the prediction: shared task pressures may rebuild some computational organizations while leaving substrate-bound or historically contingent properties behind.

Representational convergence illustrates the restraint required. Huh et al. (2024) report convergence across separately trained models, while Gröger et al. (2026) show that width, depth, and layer selection confound global spectral comparisons. Calibrated local-neighbourhood similarity survives more readily than global convergence. That evidence may warrant a bounded projection about local representational structure. It doesn't establish that training maintains cognition as a cluster, still less that it corrects perturbations to preserve such a cluster.

The word *homeostatic* should be reserved for perturbation-sensitive corrective control. Stability, causal ordering, effective maintenance, and corrective control are distinct world-side commitments. Evidence that a pattern recurs can support stability; intervention evidence may support a causal ordering; neither alone demonstrates a controller. The strongest commitment established in the cases examined here is bounded causal organization in Hu et al., not corrective control of cognition.

8 When the framework would fail

The proposal has two failure modes. Empirically, it fails to earn its complexity if capacities, organization, memory, and agency reliably co-vary across tasks, model families, and changes from checkpoint to scaffold. Under those conditions, the bearer distinctions would add little predictive value and the broader binary question would be better posed than this paper allows.

Methodologically, the framework fails if every live disagreement ends with both sides being right relative to their purposes. Purpose selects a target; it doesn't guarantee the projection. The schema has to return failed or unlicensed inferences in some cases. It does so here: Nefdt's claim is defensible for a restricted text-only bearer but unsupported as a bearer-independent placement; Kosinski's broader theory-of-mind transfer fails the available variation test; Hu's organizational claim succeeds within declared bounds; and architecture-to-phenomenology remains unlicensed.

That distribution of verdicts is the point. Pluralism doesn't mean that anything goes. It means that source–target relations are assessed separately and can be integrated only after their warrants and limits are known.

9 Conclusion

A disciplined cognitive attribution to an LLM proceeds in six steps: (1) name the bearer and target population; (2) distinguish the observed source from the projected target; (3) declare the conditions, task-preserving transformations, and timescale; (4) state the warrant, tolerance, and prospective failure condition; (5) report the result and its transfer limit; and (6) broaden, restrict, split, suspend, or retire the claim without post hoc rescue.

This procedure doesn't show that LLMs possess cognition, lack it, or should never be classified together. It shows why a verdict about *LLMs* can't be read directly from evidence about an unspecified bearer. The base checkpoint, session, deployed system, and scaffold are related objects, not interchangeable units of attribution.

Nefdt is right that language and cognition can come apart, but wrong to treat the resulting quadrant as an unindexed placement of LLMs as such. Once the bearer is declared, the missing quadrant either describes a restricted text-only system or becomes a defeasible empirical hypothesis about integration in richer deployments. LLMs are boundary phenomena because they force familiar cognitive projections to declare where, and why, they stop.

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